

2 goals: ① What is Statistical learning?
② Prob/stat review

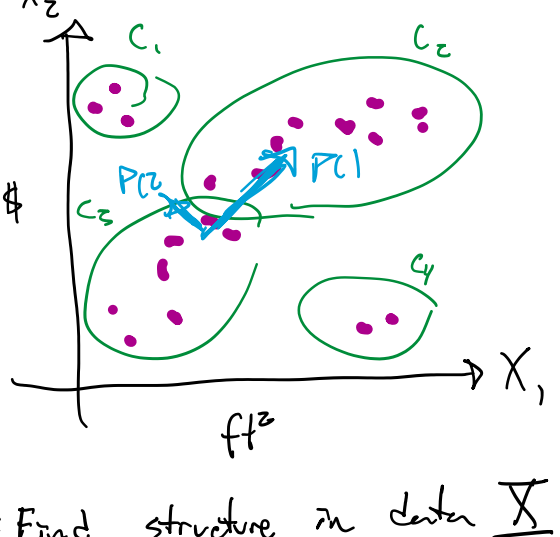
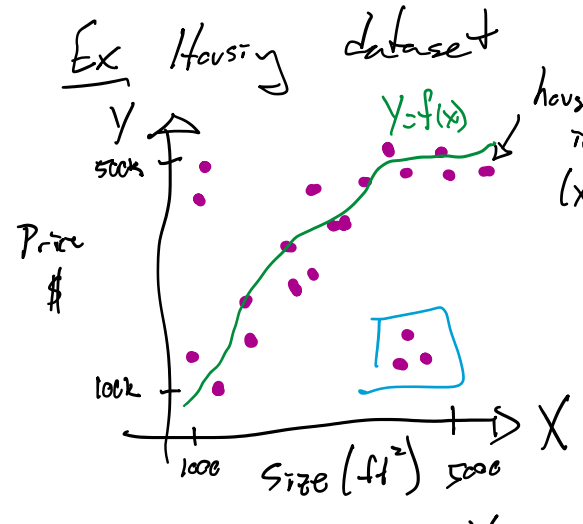
[Ch1]

What is SL? Basically, application of stat theory to machine learning.
2000s 60s in CS

2 classic categories of SL: ① Supervised learning DSL 423
② Unsupervised learning DSL 430

Very well dev & quantifiable learning vs

Unsupervised learning



Goal: Relate X & Y via predictor/feature/indep variable random $Y = f(X) + \epsilon$ error (X limitation)

Goal: Find structure in data $X = \{x_1, x_2\}$

2 main types: ① Clustering (K-means) ② Dimension reduction (PCA)

End goal: Simplify messy data for future analysis

Fuzzy & unstructured

End goal: Approx $f(x)$ via $\hat{f}(x)$ to

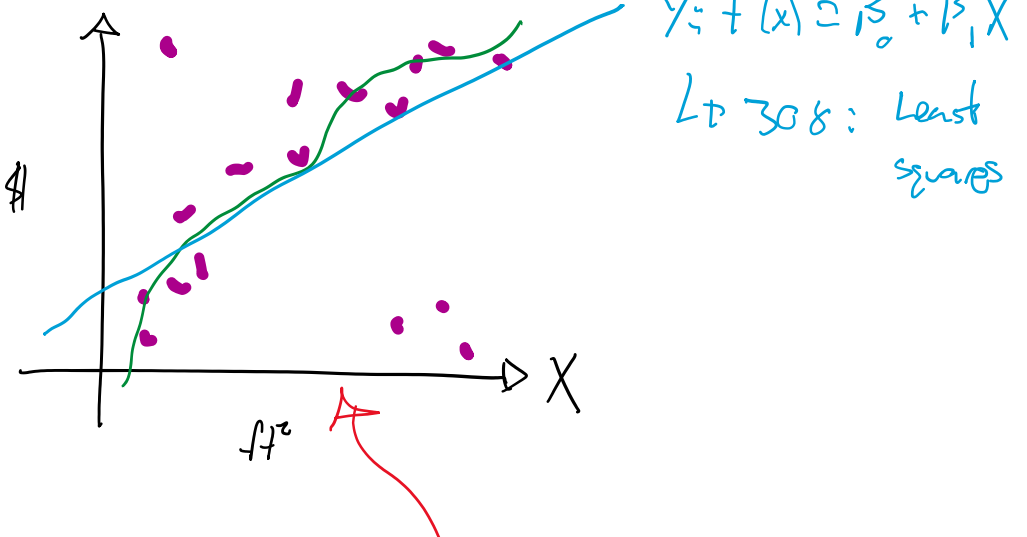
ML ① Predict on new data

SL ② Statistical inference for understanding & confidence

Intro to SL & supervised learning (Ch2)

X-Y relationship to analyze:

$$Y = f(X) + \epsilon$$



$$Y: \hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 X$$

Least squares

Notes: ① X can be 1+ dimensional vector written $X = (x_1, x_2, \dots, x_p)$

& x_i 's can be diff # predictors variable types (bool, quant, category, text, ordinal)

② Y classes: (i) $-\infty < Y < \infty$, regression problem Ch3 Linear reg.

(ii) $Y \in \{\theta_1, \theta_2, \dots, \theta_c\}$, classification Ch4 Logistic reg.

Goal: Approx $f(x) \approx \hat{f}(x)$ by fitting $\hat{f}$ to existing data

Ex Assume $f(x)$ is linear, $f(x) = \beta_0 + \beta_1 X$, then

linear regression gives $f(x) \approx \hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 X$

2 key stat results on $f(x) \approx \hat{f}(x)$

① Reducible & irreducible error

② Bias-variance trade off

Theorem If $Y = f(X) + \epsilon$ for X, Y, f, ϵ above, then

$$E[(f(X) - \hat{f}(X))^2] = (f(X) - \hat{f}(X))^2 + \text{Var}(\epsilon)$$

 Expected value (input) output (fixed values) random variable with non-zero (irreducible error) Variance (reducible error) (control)

Bias idea: $Y \approx \hat{Y}$ approx can be improved by choosing better $\hat{f}$ (smaller red error), but ϵ error cannot be reduced (irred error)

Discrete Probability crash course (Stat 245 v. fast)

Ex Roll a die



fair

Sample space S is the set of all possible outcomes.

Ex $S = \{1, 2, 3, 4, 5, 6\}$

An event $E \subseteq S$ is any subset of S .

Ex $E = \{2, 3\}$, $E_{\text{even}} = \{2, 4, 6\}$

The probability of event E is $P(E) = \frac{|E|}{|S|}$

Ex $P(E) = \frac{1}{6}$, $P(E_{\text{even}}) = \frac{3}{6} = \frac{1}{2}$.

Probability rules ① $0 \leq P(E) \leq 1$

② $P(S) = 1$

③ $P(E^c) = 1 - P(E)$

④ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

⑤ If A & B are mutually exclusive, $A \cap B = \{\} = \emptyset$, then

$$P(A \cup B) = P(A) + P(B)$$

The conditional probability of A given B occurred is

$$P(A) = P(A|B) = \frac{P(A \cap B)}{P(B)} \left(= \frac{|A \cap B|/|S|}{|B|/|S|} \right)$$



If A & B are independent, then

$$P(A \cap B) = P(A) \cdot P(B)$$

$$\text{Ex } P(E_2 | E_{\text{even}}) = \frac{P(E_2 \cap E_{\text{even}})}{P(E_{\text{even}})} = \frac{1}{3}$$

A (discrete) random variable X assigns a number to each outcome.

Ex Die roll already has X in $\{1, \dots, 6\}$

Ex Flip 2 coins, count # of heads and X in $\{0, 1, 2\}$

A probability mass function assigns a prob to each X possibility,

$$p(x) = P(X = x)$$

function notation (prob notation)

Ex Die, $p(x) = \frac{1}{6}$ all x

Ex 2 coins, $p(0) = \frac{1}{4}$

$p(1) = \frac{1}{2}$

$p(2) = \frac{1}{4}$

HH

HT

TH

TT

The expected value of X is

$$E[X] = \sum_x x p(x)$$

Long term avg of X

$$\text{Ex Die, } E[X] = \sum_{x=1}^6 x p(x) = \sum_{x=1}^6 x \left(\frac{1}{6}\right) = \frac{1}{6} (1+2+\dots+6)$$

$$= \frac{1}{6} \frac{6 \cdot 7}{2} = \frac{7}{2} = 3.5$$

means? Long term avg

Generalization:

$$E[g(x)] = \sum_x g(x) p(x)$$

any fun of d.r.v. X

The variance of X is

$$\text{Var}(X) = E[(X - \mu)^2], \mu = E[X]$$

Avg of dist from X to middle

$$\text{Ex Die, } \text{Var}(X) = E[(X - \mu)^2] = E[(X - 3.5)^2]$$

$$= \sum_{x=1}^6 (x - 3.5)^2 p(x) = \frac{1}{6} \sum_{x=1}^6 (x - 3.5)^2$$

$$\text{Then } \text{Var}(X) = E[X^2] - (E[X])^2$$

Linearity prop: $E[cX] = cE[X]$

$$E[X_1 + X_2] = E[X_1] + E[X_2]$$

$$\text{Proof } E[(X - \mu)^2] = E[X^2 - 2X\mu + \mu^2] = E[X^2] - 2\mu E[X] + \mu^2$$

$$= E[X^2] - (E[X])^2$$